

Samata Sainik Dal (SSD)

Digital Command, Cadre Enlistment & Public Credential Verification Platform

SYSTEM ARCHITECTURE
& WORKFLOW BLUEPRINT
V2.0

Executive Summary

Founded in 1927 by **Dr. B. R. Ambedkar**, Samata Sainik Dal (Army of Equality) is an organized social defense cadre dedicated to equality, constitutional rights, and community welfare. This digital platform modernizes national operations by interconnecting **public citizen outreach**, **multi-tiered enlistment & review workflows**, **active cadre command**, **fraud-proof QR verification**, and **statutory 80G treasury management** on a resilient, high-availability architecture.

1 High-Level Architecture Topology

LAYER 1 — CLIENT PRESENTATION LAYER (RESPONSIVE WEB)

VANILLA HTML5 / MODERN CSS / ES6+

Public Web Portal

index, about, wings, contact

Cadre Member Portal

member-portal.html & .js

Central Command Admin

admin.html & admin.js

LAYER 2 — API GATEWAY & SECURITY PERIMETER

EXPRESS 4.21 / HELMET / CORS

JWT & JWKS Auth

middleware/auth.js

Hierarchical RBAC

middleware/rbac.js

Territorial Bounds

enforceJurisdiction()

Rate Limiter & Upload

Multer + Express Limit

LAYER 3 — CORE BUSINESS DOMAIN SERVICES (RESTFUL APIS)

MODULAR EXPRESS ROUTERS

Enlistment & Approvals

/api/membership

Active Cadre & Cards

/api/members

QR Verification Engine

/api/verify

Treasury & 80G Receipts

/api/donations

Territory & Chapters

/api/chapters

Gazette & Events CMS

/api/news, /api/events

LAYER 4 — RESILIENT DUAL-ENGINE PERSISTENCE TIER

ZERO-DOWNTIME FAILOVER

Primary: Supabase Cloud Database

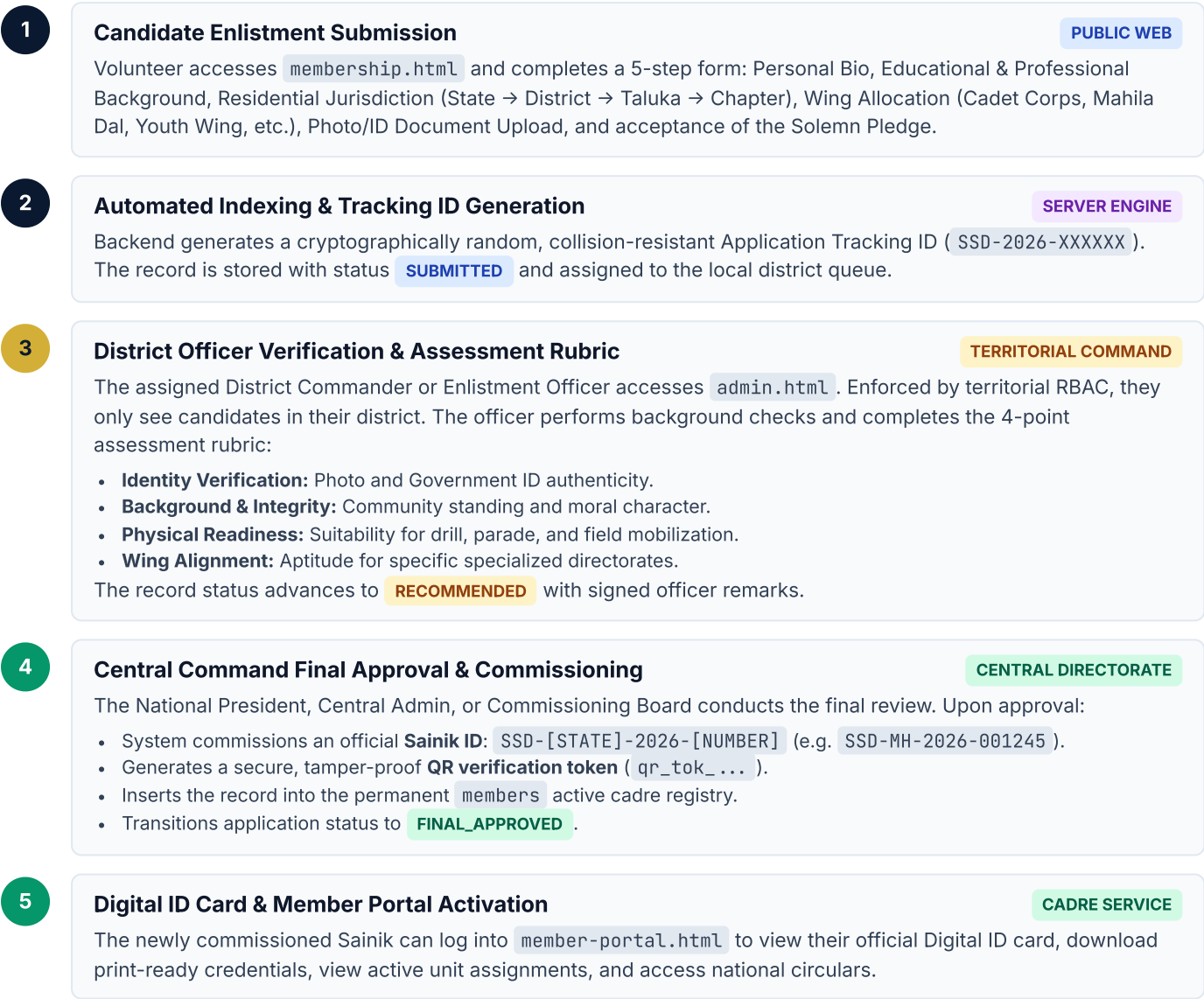
PostgreSQL + Relations + @supabase/server SDK

Failover: Embedded Transactional Store

In-Memory Maps + Auto-Sync data_store.json

2 End-to-End Enlistment & Commissioning Workflow

This pipeline transitions a volunteer from initial digital application to a fully commissioned, uniformed Sainik with a unique alphanumeric ID and cryptographic QR credentials.



3 Public QR Verification & Anti-Fraud Shield

Designed for rapid verification by civil authorities, police, event organizers, and the public during rallies and disaster response operations.

5-Tier High-Availability Lookup Pipeline

When a QR code is scanned or an identifier is submitted on `verify.html`:

- URL Normalization:** Strips host and query parameters if a native mobile camera app returns the full browser link.
- Tier 1 — Backend API:** Queries `GET /api/verify/:id`.
- Tier 2 — Card API Fallback:** Calls `GET /api/members/card/:id` if primary endpoint is busy.
- Tier 3 — Direct Supabase Client:** Direct browser-to-database query via Supabase JS SDK.
- Tier 4 — Local Storage Cache:** Reads cached offline credential records.
- Tier 5 — Algorithmic Validation:** Validates checksum patterns for emergency validation.

Supported Identification Tokens

The verification engine accepts any of the following parameters:

- Official Sainik ID:** `SSD-MH-2026-001245`
- Application Reference:** `SSD-2026-8F42K7`
- 10-Digit Mobile Number:** `9822341927`
- Cryptographic QR Token:** `qr_tok_...`

PUBLIC VERIFICATION DISPLAY BADGE

SAMATA SAINIK DAL

✓ OFFICIALLY VERIFIED ACTIVE CADRE

Sainik ID:	SSD-MH-2026-001245
Full Name:	Subhash V. Meshram
Rank & Role:	Cadet Sainik (Drill & Guard)
Wing:	Central Cadet Corps
Jurisdiction:	Nagpur Central Unit, Maharashtra
Mobile:	+91 98223 [MASKED]
Date of Birth:	../../1992 [PRIVACY SHIELD]
Commission Date:	15 August 2026

Digitally certified by Central Command Cryptographic Authority.

Data Privacy Safeguard: Full government identification numbers, resident street addresses, and raw mobile numbers are masked to prevent identity theft.

4 Treasury, Razorpay & 80G Tax Exemption Workflow

Processes voluntary public contributions with automated tax exemption receipts under Section 80G of the Indian Income Tax Act.

1

Donor Order Initiation

PUBLIC CLIENT → SERVER API

Donor specifies amount, cause (General Welfare, Relief Fund, Training Academy), PAN number, and contact details. Server calls `razorpay.orders.create()` to register an immutable transaction order.

2

Razorpay Universal Checkout

SECURE PAYMENT GATEWAY

Browser launches Razorpay's standard checkout dialog supporting UPI (Google Pay, PhonePe, Paytm), Credit/Debit Cards, NetBanking, and Wallets.

3

Cryptographic Signature Verification

SERVER SECURITY

Payment response is sent to `POST /api/donations/verify`. The backend generates an HMAC-SHA256 digest using the secret key and ensures the signature matches before registering the donation as completed.

4

Automated 80G Receipt Issuance

TREASURY LEDGER

System generates an official sequential receipt number (`SSD-REC-2026-XXXXXX`). The donor instantly views and downloads a signed, printable donation certificate featuring SSD registration numbers and 80G tax exemption clauses.

Treasury Compliance Data Model

Field	Type	Description & Regulatory Purpose
<code>order_id</code>	VARCHAR(100) UNIQUE	Razorpay Server-generated reference identifier.
<code>payment_id</code>	VARCHAR(100) UNIQUE	Bank confirmation token for anti-money laundering reconciliation.
<code>pan</code>	VARCHAR(20)	Permanent Account Number for statutory Form 10BD income tax filing.
<code>receipt_number</code>	VARCHAR(100) UNIQUE	Sequential audited receipt number (<code>SSD-REC-2026-000001</code>).
<code>is_80g_eligible</code>	BOOLEAN	Flags exemption status under Section 80G of the Income Tax Act.

5 Role-Based Access Control (RBAC) & Territorial Scope

The platform enforces granular administrative separation between strategic central command and field operations.

Role Identifier	Administrative Scope	Permissions & Capabilities
super_admin	NATIONAL UNLIMITED	Full platform control: System settings, user role provisioning, database migrations, audit trail access, and final approval override.
central_admin	NATIONAL COMMAND	Central Executive authority: Commissioning recommended candidates, appointing state leadership, and publishing official gazettes.
finance_admin	NATIONAL TREASURY	Financial ledger management: Reconciliation of Razorpay collections, 80G tax receipt audits, and expenditure reporting.
state_official	SINGLE STATE BOUNDARY	State Directorate: Oversees district units within their assigned state (e.g. Maharashtra), manages state events, and reviews escalations.
district_official	SINGLE DISTRICT BOUNDARY	District Command: Inspects candidate documents, conducts assessment rubric reviews, and forwards applications with recommendation flags.
enlistment_officer	ENLISTMENT QUEUE ONLY	Focused verification officer: Reviews pending submissions and verifies applicant ID credentials. Restricted from treasury and system admin.

6 Immutable System Audit Trail

Zero-Tamper Activity Logging Architecture

Every administrative state transition (application approval, status rejection, user role alteration, or payment verification) triggers an append-only log entry in `audit_logs` and `approval_actions`. Each record logs:

Actor Fingerprint	State Transition	Timestamp & Reason
User ID, Role, IP Address, User Agent	Previous Status → New Status	UTC Timestamp + Officer Remarks

7 Relational Data Model & Core Schema

Normalized relational design connecting geographic hierarchies, workflow states, and financial transactions.

membership_applications	
id (PK)	VARCHAR(64) — SSD-2026-8F42K7
full_name	VARCHAR(255) NOT NULL
mobile, email	VARCHAR(30), VARCHAR(255)
state_id, district_id	FK to states and districts
status	SUBMITTED, RECOMMENDED, FINAL_APPROVED
documents_json	JSONB — Photo & Proof URLs

members (Active Cadre)	
id (PK)	VARCHAR(64) — UUID
sainik_id (UK)	VARCHAR(64) — SSD-MH-2026-001245
application_id	FK to membership_applications
qr_token (UK)	VARCHAR(255) — Cryptographic token
designation	VARCHAR(150) — Cadet Sainik, etc.
status	ACTIVE, SUSPENDED, RETIRED

approval_actions (Audit Trail)	
id (PK)	VARCHAR(64)
application_id	FK to membership_applications
official_id, role	Actor identification & scope
action	RECOMMEND, FINAL_APPROVE, REJECT
remarks	Assessment rubric notes & findings

donations & receipts	
order_id	Razorpay order reference token
payment_id	Gateway completion transaction ID
amount, pan	NUMERIC(12,2), Income tax PAN
receipt_number	SSD-REC-2026-XXXXXX
is_80g_eligible	Statutory deduction qualification

8 Resilience & Reliability Guarantee

Dual-Layer Storage Cloud Supabase PostgreSQL synced with local in-memory transactional cache ensures 0% downtime during network blackouts.	Automated Tests 17 end-to-end automated test suites validate RBAC, enlistment workflows, QR lookups, and Razorpay HMAC signature verification.	Security Compliance Zero plain-text password storage, bcrypt hashing, rate limiting, and privacy masking of personal identification data.
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